Key Insights

Mobile App User Engagement and Subscription Analysis

Engagement Patterns

- **Operating system:** Android recorded significantly more interactions than iOS in the analyzed dataset.
- **Device model:** Xiaomi Mi 11 was the leading device model by interaction count.
- **Event type:** Click was the most frequent event type, followed by share and view.
- **Event target:** The checkout button was the most-engaged event target.

Subscription Patterns

- The subscription distribution was close to evenly split, with slightly more than half of the records/users classified as subscribed in the project documentation.
- The dashboard compares subscription rates by country and operating system to help identify differences across these categories.

Numerical Relationships

- Battery level, memory usage, event value, session duration, and user age showed weak linear correlations with one another in the reported exploratory analysis.
- The dashboard includes a memory-usage-versus-session-duration scatter plot by device model and compares session duration, battery level, and event value across categories.

Dashboard Coverage

- Summary metrics include approximately 99K total interactions, average session duration of approximately 1.80K seconds, average memory usage of approximately 2.06K MB, and subscription rate.
- Visuals cover interactions by event type and operating system, top device models, top countries, subscription rates by country and operating system, average session duration by operating system, average event value by event type, battery level by network type, and an event-type/event-target matrix.

Interpretation and Limitations

- These findings describe the analyzed dataset; they should not be interpreted as proof of cause and effect.
- Missing-value imputation may affect the variability of numerical and categorical fields.
- The documented analysis has limited segmentation by city and age.
- The project documentation does not provide statistical significance tests or evidence that interaction patterns predict future subscriptions.

Potential Next Steps

- Add time-based trend and cohort analysis.
- Explore user segments by city and age where sample sizes and data quality allow.

- Evaluate conversion prediction using appropriate validation methods.
- Explore nonlinear relationships and, if available, combine interaction data with marketing or revenue data.